

Raghav Chaturvedi

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📅 Date of Birth: 25 Sep 1997 🇮🇳 Nationality: Indian



Professional Summary

Physicist and scientific software engineer with 8 years of experience building and deploying numerical simulation tools for quantum systems. Skilled in Python (NumPy, SciPy, scikit-learn, pandas), C++, Git-based collaborative development, and high-performance computing (MPI, Slurm). Track record of turning research-grade scripts into reusable, documented software used by other scientists, and of applying machine learning to accelerate computationally expensive physics workflows. PhD in Theoretical/Computational Physics (Magna Cum Laude) from IFW/TU Dresden; postdoc at JMU Würzburg.

Technical Skills

- **Programming & Scientific Computing:** Python, NumPy, SciPy, pandas, scikit-learn, PyTorch, TensorFlow, C++, Kwant, Qiskit, Bash scripting, Git, High-Performance Computing (MPI, Slurm), LaTeX.
- **Numerical Methods:** Linear algebra, eigenvalue problems, matrix diagonalization, scattering-matrix methods, ODE solvers (RK4, Euler), Monte Carlo simulations, finite-size analysis, optimization.
- **Quantum Computing & Information:** Quantum circuit implementation and basic algorithms (Qiskit); conceptual understanding of quantum error correction (stabilizer codes, Shor code) and quantum key distribution (QKD), from graduate coursework in quantum information theory and quantum communication networks.
- **Software:** Mathematica, Matlab, Origin, Inkscape.
- **Languages:** English (Proficient), German (B1 – TELC certified, currently learning B2).

Key Projects

- Quantum Transport Simulation Library** | *Python, NumPy, Kwant, Git* 2024–present
 - **Problem:** Simulation work in the group lived in fragmented, one-off scripts with no shared structure or reuse.
 - **Built:** Modular Python library providing reusable device geometry classes, multi-terminal current solvers, and batch parameter sweep utilities. Clean API, typed schemas, version control, documentation.
 - **Impact:** Adopted by 2–3 researchers; reduces multi-day scripting to a few function calls. To be released publicly with an upcoming paper.
- ML Pipeline for Automated Parameter Selection** | *scikit-learn, pandas, Random Forest* 2025–present
 - **Problem:** A characteristic length scale had to be identified manually for each of thousands of non-Hermitian tight-binding chain configurations – slow and inconsistent.
 - **Built:** Two-stage algorithm: logistic regression to flag valid cases; Random Forest regressor (500 trees) trained on 17,000+ non-Hermitian chain Hamiltonians to predict that length scale. Packaged as an installable Python library with a command-line interface and a pytest test suite, so any matrix can be evaluated without rerunning the training pipeline.
 - **Impact:** R-squared of 0.88–0.98; more accurate than manual estimation. Turned a manual, ad hoc step into a reusable command-line tool usable by other researchers.
- Simulation-to-Hardware: Theory Lead on a Fabricated Quantum Device** | *Python, Kwant* 2020–2024
 - **Role:** Theory PhD on a joint theory–experiment project. Built numerical simulations of the device from scratch, initiated and organised regular cross-group meetings, and kept simulation and fabrication aligned throughout.
 - **Built:** Device simulation matched to experimental geometry; computed transport properties and invariants that served as predictions for the fabrication team.
 - **Impact:** Simulations preceded and guided device fabrication; predictions validated against measurements. Published in *Nature Physics* 20, 395–401 (2024) as joint first author. Experimental group subsequently built a graphene-based follow-up device; co-authored study under review in *Nature Communications*.

Certifications

- Machine Learning Specialization (Stanford University – Coursera): supervised/unsupervised learning, regression, classification, clustering; Python, NumPy, scikit-learn. [\[Link\]](#)
- Deep Learning Specialization (DeepLearning.AI – Coursera): DNNs, CNNs, RNNs, optimization, regularization in TensorFlow. [\[Link\]](#)
- Introduction to HPC and Parallel Computing (University of Colorado Boulder – Coursera): MPI, OpenMP, GPU basics, Slurm job submission. [\[Link\]](#)

Management & Communication Skills

- **Team leadership:** Founded and led Utkalp, a science-education platform, coordinating a remote team of 12 (writers, editors, designers, social-media managers) across multiple cities; grew it to 3,000+ subscribers.
- **Mentoring & teaching:** Mentored internship students through to publication; taught undergraduate physics laboratory for two semesters (Grundpraktikum III).
- **Grant writing:** Co-wrote a successful research funding proposal for the ct.qmat Cluster of Excellence, securing support for the current postdoctoral position.
- **Communication:** Presented research to international audiences across India, the USA, Canada, and Germany – including contributed talks at the **APS March Meeting** and **DPG meetings**, an invited international seminar, and posters at international conferences and schools.
- **Peer review:** Reviewer for **Cell Press Network**, refereeing original condensed matter research.

Work Experience

Julius-Maximilians-University (JMU) <i>Postdoctoral Researcher – Theoretical Condensed Matter Physics</i>	Würzburg, Germany Oct 2024–present
• Built and maintains the group's core simulation library in Python; adopted by multiple researchers (see Projects). • Designed and deployed an ML pipeline to automate a key analysis step across thousands of simulations, replacing manual computation (see Projects). • Leading theory modelling for an ongoing experimental collaboration on semiconductor spin-transport devices – translating raw experimental observations into validated numerical models.	
Leibniz Institute for Solid State and Materials Research (IFW) <i>Doctoral Researcher – Theoretical Condensed Matter Physics</i>	Dresden, Germany Oct 2020–Feb 2025
• Thesis: <i>Non-Hermitian topology of transport in quantum Hall systems</i>. Grade: Magna Cum Laude. • Developed and validated computational models of quantum devices across 1D–3D geometries; computed physical observables matched to experimental measurements (see Projects).	
Utkalp <i>Founder & Team Lead – Science EdTech YouTube Channel</i>	India (remote) 2020–2023
• Founded and grew a multi-lingual science-education YouTube channel (Hindi, English, regional Indian languages); led the team (see Management & Communication Skills).	

Education

Leibniz Institute / TU Dresden <i>Doctor of Philosophy (Theoretical Condensed Matter Physics) – Magna Cum Laude</i>	Dresden, Germany Oct 2020–Feb 2025
Indian Institute of Science Education and Research (IISER) <i>BS–MS Dual Degree in Physics (CGPA 8.77/10 – 3rd in a class of 140)</i>	Trivandrum, India Aug 2015–Aug 2020

Awards & Fellowships

- KVPY Scholar: Fellowship awarded by Indian Institute of Science, Bangalore to top 2000 students nationally.
- DST-INSPIRE Scholar: Fellowship awarded by the Government of India to the top 1% of students.
- Indian Academy of Sciences Summer Research Fellow (4 fellowships, 2016–2019): competitively selected for fully-funded research placements at leading Indian institutes.

Selected Publications

- *arXiv:2607.05900* (2026) – Machine learning prediction of the convergence criterion for a topological invariant of finite non-Hermitian chains. **First author.** [\[Link\]](#)
- *Nature Physics* 20, 395–401 (2024) – Non-Hermitian topology in a multi-terminal quantum Hall device. **Joint first author.** [\[Link\]](#)
- *Phys. Rev. B* 111, 245424 (2025) – Non-Hermitian topology of transport in graphene quantum Hall phases. [\[Link\]](#)
- 2 further preprints on arXiv (2024–2026). Full list: [Google Scholar](#).

References

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